

The texvc package*

Moritz Schubotz
moritz.schubotz@fiz-karlsruhe.de

2025/11/15

Abstract

This package provides all¹ LaTeX command available in MediaWiki. This includes several packages like amsmath, and adds some specific commands such as `\Reals`.

1 Provided Macros

1.1 Arrows

The first group of MediaWiki custom command (`other_delimiters2`) defines short hand notations for some arrows.

`\darr` Short hand notation for arrow $\downarrow$.

`\dArr` Short hand notation for arrow $\Downarrow$.

`\Darr` Short hand notation for arrow $\Downarrow$.

`\lang` Short hand notation for arrow $\langle$.

`\rang` Short hand notation for arrow $\rangle$.

`\uarr` Short hand notation for arrow $\uparrow$.

`\uArr` Short hand notation for arrow $\Uparrow$.

`\Uarr` Short hand notation for arrow $\Uparrow$.

1.2 Literals

The second group of MediaWiki custom commands (`other_litereals3`) defines short hand notations for some literals.

`\C` Short hand notation for literal $\mathbb{C}$. *This command is deprecated.*

`\H` Short hand notation for literal $\mathbb{H}$. *This command is deprecated.*

`\N` Short hand notation for literal $\mathbb{N}$.

`\Q` Short hand notation for literal $\mathbb{Q}$.

*This document corresponds to texvc v1.3, dated 2025/11/15.

¹The command `\or` is only available if custom code is copied into your $\LaTeX$ -file. See page 10 for details.

`\R` Short hand notation for literal $\mathbb{R}$.
`\Z` Short hand notation for literal $\mathbb{Z}$.
`\alef` Short hand notation for literal $\aleph$.
`\alefsym` Short hand notation for literal $\aleph$.
`\Alpha` Short hand notation for literal A .
`\and` Short hand notation for literal $\wedge$. *This command is deprecated.*
`\ang` Short hand notation for literal $\angle$. *This command is deprecated.*
`\Beta` Short hand notation for literal B .
`\bull` Short hand notation for literal $\bullet$.
`\Chi` Short hand notation for literal X .
`\clubs` Short hand notation for literal $\clubsuit$.
`\cnums` Short hand notation for literal $\mathbb{C}$.
`\Complex` Short hand notation for literal $\mathbb{C}$.
`\Dagger` Short hand notation for literal $\ddagger$.
`\diamonds` Short hand notation for literal $\diamond$.
`\Doteq` Short hand notation for literal $\doteq$.
`\doublecap` Short hand notation for literal $\mho$.
`\doublecup` Short hand notation for literal Ψ .
`\empty` Short hand notation for literal $\emptyset$.
`\Epsilon` Short hand notation for literal E .
`\Eta` Short hand notation for literal H .
`\exist` Short hand notation for literal $\exists$.
`\ge` Short hand notation for literal $\geq$.
`\gggtr` Short hand notation for literal $\gggtr$.
`\hAar` Short hand notation for literal $\Leftrightarrow$.
`\harr` Short hand notation for literal $\leftrightarrow$.
`\Harr` Short hand notation for literal $\Leftrightarrow$.
`\hearts` Short hand notation for literal $\heartsuit$.
`\image` Short hand notation for literal $\Im$.
`\infin` Short hand notation for literal ∞ .
`\Iota` Short hand notation for literal I .
`\isin` Short hand notation for literal $\in$.
`\Kappa` Short hand notation for literal K .

`\larr` Short hand notation for literal $\leftarrow$.
`\Larr` Short hand notation for literal $\Leftarrow$.
`\lArr` Short hand notation for literal $\Lleftarrow$.
`\le` Short hand notation for literal $\leq$.
`\lrarr` Short hand notation for literal $\leftrightarrow$.
`\Lrarr` Short hand notation for literal $\Leftrightarrow$.
`\lrArr` Short hand notation for literal $\Leftrightarrow$.
`\Mu` Short hand notation for literal $\mathbb{M}$.
`\natnums` Short hand notation for literal $\mathbb{N}$.
`\ne` Short hand notation for literal $\neq$.
`\Nu` Short hand notation for literal $\mathbb{N}$.
`\O` Short hand notation for literal $\emptyset$.
`\omicron` Short hand notation for literal o .
`\Omicron` Short hand notation for literal O .
`\or` Short hand notation for literal $\vee$. *This command is deprecated.*
`\part` Short hand notation for literal ∂ . *This command is deprecated.*
`\plusmn` Short hand notation for literal $\pm$.
`\rarr` Short hand notation for literal $\rightarrow$.
`\Rarr` Short hand notation for literal $\Rightarrow$.
`\rArr` Short hand notation for literal $\Rightarrow$.
`\real` Short hand notation for literal $\Re$.
`\reals` Short hand notation for literal $\mathbb{R}$.
`\Reals` Short hand notation for literal $\mathbb{R}$.
`\restriction` Short hand notation for literal $\upharpoonright$.
`\Rho` Short hand notation for literal $\mathbb{P}$.
`\sdot` Short hand notation for literal $\cdot$.
`\sect` Short hand notation for literal $\S$.
`\spades` Short hand notation for literal $\spadesuit$.
`\sub` Short hand notation for literal $\subset$.
`\sube` Short hand notation for literal $\subseteq$.
`\supe` Short hand notation for literal $\supseteq$.
`\Tau` Short hand notation for literal $\mathbb{T}$.
`\thetasym` Short hand notation for literal ϑ .

`\varcoppa` Short hand notation for literal $\wp$.

`\weierp` Short hand notation for literal $\wp$.

`\Zeta` Short hand notation for literal Z .

2 Deprecations

According to the decision of *Wikimedia Community User Group Math*² the following macros have been deprecated³:

1. `$`
2. `%`
3. `\and`
4. `\or`
5. `\part`
6. `\ang`
7. `\C`
8. `\H`
9. `\bold`
10. `\Bbb`

The commands 1,2,4,9,10 have never been part of this package, but were available from within Wikipedia.

²<https://meta.wikimedia.org/w/index.php?oldid=19705444>

³See <https://phabricator.wikimedia.org/T197842> for the discussion.

3 Dependencies

MediaWiki uses LaTeX macros from the following packages in addition to the standard LaTeX packages:

- amsmath
- amsfons
- amssymb
- color
- babel
- teubner
- eurosym
- cancel
- bbold
- mhchem
- stix
- arcs

This LaTeX package requires those packages automatically, so that all commands that are usable on MediaWiki are also usable in LaTeX. For a list of the provided commands, see Section A. In addition, the MediaWiki help page for math (see <https://en.wikipedia.org/wiki/WP:Math>) provides additional details on their usage.

4 Implementation

`\darr` This macro does the following replacement.

```
1 \newcommand{\darr}{\downarrow}
```

`\dArr` This macro does the following replacement.

```
2 \newcommand{\dArr}{\Downarrow}
```

`\Darr` This macro does the following replacement.

```
3 \newcommand{\Darr}{\Downarrow}
```

`\lang` This macro does the following replacement.

```
4 \newcommand{\lang}{\langle}
```

`\rang` This macro does the following replacement.

```
5 \newcommand{\rang}{\rangle}
```

`\uarr` This macro does the following replacement.

```
6 \newcommand{\uarr}{\uparrow}
```

`\uArr` This macro does the following replacement.

```
7 \newcommand{\uArr}{\Uparrow}
```

`\Uarr` This macro does the following replacement.

```
8 \newcommand{\Uarr}{\Uparrow}
```

`\C` This macro does the following replacement.

```
9 %\newcommand{\C}{\mathbb{C}}
```

`\H` This macro does the following replacement.

```
10 \renewcommand{\H}{\mathbb{H}}
```

`\N` This macro does the following replacement.

```
11 \newcommand{\N}{\mathbb{N}}
```

`\Q` This macro does the following replacement.

```
12 \newcommand{\Q}{\mathbb{Q}}
```

`\R` This macro does the following replacement.

```
13 \newcommand{\R}{\mathbb{R}}
```

`\Z` This macro does the following replacement.

```
14 \newcommand{\Z}{\mathbb{Z}}
```

`\alef` This macro does the following replacement.

```
15 \newcommand{\alef}{\aleph}
```

`\alefsym` This macro does the following replacement.

```
16 \newcommand{\alefsym}{\aleph}
```

`\Alpha` This macro does the following replacement.

```
17 \newcommand{\Alpha}{\mathrm{A}}
```

`\and` This macro does the following replacement.

```
18 \renewcommand{\and}{\land}
```

`\ang` This macro does the following replacement.

```
19 \newcommand{\ang}{\angle}
```

`\Beta` This macro does the following replacement.

```
20 \newcommand{\Beta}{\mathrm{B}}
```

`\bull` This macro does the following replacement.

```
21 \newcommand{\bull}{\bullet}
```

`\Chi` This macro does the following replacement.

```
22 \newcommand{\Chi}{\mathrm{X}}
```

`\clubs` This macro does the following replacement.

```
23 \newcommand{\clubs}{\clubsuit}
```

`\cnums` This macro does the following replacement.

```
24 \newcommand{\cnums}{\mathbb{C}}
```

`\Complex` This macro does the following replacement.

```
25 \newcommand{\Complex}{\mathbb{C}}
```

`\Dagger` This macro does the following replacement.

```
26 \newcommand{\Dagger}{\ddagger}
```

`\diamonds` This macro does the following replacement.

```
27 \newcommand{\diamonds}{\diamondsuit}
```

`\Doteq` This macro does the following replacement.

```
28 \renewcommand{\Doteq}{\doteqdot}
```

`\doublecap` This macro does the following replacement.

```
29 \renewcommand{\doublecap}{\Cap}
```

`\doublecup` This macro does the following replacement.

```
30 \renewcommand{\doublecup}{\Cup}
```

`\empty` This macro does the following replacement.

```
31 \renewcommand{\empty}{\emptyset}
```

`\Epsilon` This macro does the following replacement.

```
32 \newcommand{\Epsilon}{\mathrm{E}}
```

`\Eta` This macro does the following replacement.

```
33 \newcommand{\Eta}{\mathrm{H}}
```

`\exist` This macro does the following replacement.

```
34 \newcommand{\exist}{\exists}
```

`\ge` This macro does the following replacement.

```
35 \renewcommand{\ge}{\geq}
```

`\gggtr` This macro does the following replacement.

```
36 \renewcommand{\gggtr}{\ggg}
```

`\hAar` This macro does the following replacement.

```
37 \newcommand{\hAar}{\Leftrightarrow}
```

`\harr` This macro does the following replacement.

```
38 \newcommand{\harr}{\leftrightharrow}
```

`\Harr` This macro does the following replacement.

```
39 \newcommand{\Harr}{\Leftrightarrow}
```


`\hearts` This macro does the following replacement.

```
40 \newcommand{\hearts}{\heartsuit}
```

`\image` This macro does the following replacement.

```
41 \newcommand{\image}{\Im}
```

`\infin` This macro does the following replacement.

```
42 \newcommand{\infin}{\infty}
```

`\Iota` This macro does the following replacement.

```
43 \newcommand{\Iota}{\mathrm{I}}
```

`\isin` This macro does the following replacement.

```
44 \newcommand{\isin}{\in}
```

`\Kappa` This macro does the following replacement.

```
45 \newcommand{\Kappa}{\mathrm{K}}
```

`\larr` This macro does the following replacement.

```
46 \newcommand{\larr}{\leftarrow}
```

`\Larr` This macro does the following replacement.

```
47 \newcommand{\Larr}{\Leftarrow}
```

`\lArr` This macro does the following replacement.

```
48 \newcommand{\lArr}{\Leftarrow}
```

`\le` This macro does the following replacement.

```
49 \renewcommand{\le}{\leq}
```

`\lrarr` This macro does the following replacement.

```
50 \newcommand{\lrarr}{\leftrightarrow}
```

`\Lrarr` This macro does the following replacement.

```
51 \newcommand{\Lrarr}{\Leftrightarrow}
```

`\lrArr` This macro does the following replacement.

```
52 \newcommand{\lrArr}{\Leftrightarrow}
```

`\Mu` This macro does the following replacement.

```
53 \newcommand{\Mu}{\mathrm{M}}
```

`\natnums` This macro does the following replacement.

```
54 \newcommand{\natnums}{\mathbb{N}}
```

`\ne` This macro does the following replacement.

```
55 \renewcommand{\ne}{\neq}
```

`\Nu` This macro does the following replacement.

```
56 \newcommand{\Nu}{\mathrm{N}}
```

`\O` This macro does the following replacement.

```
57 \renewcommand{\O}{\emptyset}
```

`\omicron` This macro does the following replacement.

```
58 \newcommand{\omicron}{\mathrm{o}}
```

`\Omicron` This macro does the following replacement.

```
59 \newcommand{\Omicron}{\mathrm{O}}
```

`\or` This is a problematic macro, since it redefines the plain $\TeX$ macro `\or`. For instance, the `\thanks` command uses a custom function to determine the footnotesymbol, which relies on the availability of the `\or` command in math mode. Thus, the macro has to be defined after `\maketitle` was executed. However, there might be more commands that use `\or` used in mathmode. Thus we don't overwrite `\or` in this package. To enable the overwriting copy the code below to an appropriate position in your $\LaTeX$ -file. However, it might be easier to manually replace `\or` with `\lor` which is all what the macro above does.

```
60 %\let\oldor\or
61 %\def\or{\ifmmode\lor\else\expandafter\oldor\fi}
```

`\part` This macro does the following replacement.

```
62 \renewcommand{\part}{\partial}
```

`\plusmn` This macro does the following replacement.

```
63 \newcommand{\plusmn}{\pm}
```

`\rarr` This macro does the following replacement.

```
64 \newcommand{\rarr}{\rightarrow}
```

`\Rarr` This macro does the following replacement.

```
65 \newcommand{\Rarr}{\rightarrow}
```

`\rArr` This macro does the following replacement.

```
66 \newcommand{\rArr}{\rightarrow}
```

`\real` This macro does the following replacement.

```
67 \newcommand{\real}{\Re}
```

`\reals` This macro does the following replacement.

```
68 \newcommand{\reals}{\mathbb{R}}
```

`\Reals` This macro does the following replacement.

```
69 \newcommand{\Reals}{\mathbb{R}}
```

`\restriction` This macro does the following replacement.

```
70 \renewcommand{\restriction}{\upharpoonright}
```

`\Rho` This macro does the following replacement.

```
71 \newcommand{\Rho}{\mathrm{P}}
```

`\sdot` This macro does the following replacement.

```
72 \newcommand{\sdot}{\cdot}
```

`\sect` This macro does the following replacement.

```
73 \newcommand{\sect}{\S}
```

`\spades` This macro does the following replacement.

```
74 \newcommand{\spades}{\spadesuit}
```

`\sub` This macro does the following replacement.

```
75 \newcommand{\sub}{\subset}
```

`\sube` This macro does the following replacement.

```
76 \newcommand{\sube}{\subseteq}
```

`\supe` This macro does the following replacement.

```
77 \newcommand{\supe}{\supseteq}
```

`\Tau` This macro does the following replacement.

```
78 \newcommand{\Tau}{\mathrm{T}}
```

`\thetasy` This macro does the following replacement.

```
79 \newcommand{\thetasy}{\vartheta}
```

`\varcoppa` This macro does the following replacement.

```
80 \newcommand{\varcoppa}{\mbox{\coppa}}
```

`\weierp` This macro does the following replacement.

```
81 \newcommand{\weierp}{\wp}
```

`\Zeta` This macro does the following replacement.

```
82 \newcommand{\Zeta}{\mathrm{Z}}
```

Acknowledgements

This LaTeX package was co-funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – Project number 460135501. It is actively maintained by the *Wikimedia Community User Group Math*⁴ and the MaRDI project⁵ which uses MediaWiki as the portal for mathematical research data.

Change History

v1.0		v1.2	
General: Initial version	1	General: Document deprecations	1
v1.1		v1.3	
General: Fix bug with varcoppa, document usage of or	1	General: Expand documentation and require all packages needed to render the commands correctly. . . .	1

Index

Numbers written in *italic* refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in *roman* refer to the code lines where the entry is used.

Symbols	A	<code>\Alpha</code>	2, <u>17</u>
	<code>\alef</code>	<code>\and</code>	2, <u>15</u>
	<code>\alefsym</code>	<code>\ang</code>	2, <u>16</u>
<code>\@oldor</code>	<code>\aleph</code>	<code>\angle</code>	2, <u>19</u>
	60, 61	15, 16	19

⁴https://meta.wikimedia.org/wiki/Wikimedia_Community_User_Group_Math

⁵<https://portal.mardi4nfdi.de/>

B		$\backslash$ Harr 2, <u>39</u>	O	
$\backslash$ Beta 2, <u>20</u>		$\backslash$ harr 2, <u>38</u>	$\backslash$ O 3, <u>57</u>	
$\backslash$ bull 2, <u>21</u>		$\backslash$ hearts 2, <u>40</u>	$\backslash$ Omicron 3, <u>59</u>	
$\backslash$ bullet 21		$\backslash$ heartsuit 40	$\backslash$ omicron 3, <u>58</u>	
C		I		$\backslash$ or 3, <u>60</u>
$\backslash$ C 1, <u>9</u>		$\backslash$ ifmmode 61	P	
$\backslash$ Cap 29		$\backslash$ Im 41	$\backslash$ part 3, <u>62</u>	
$\backslash$ cdot 72		$\backslash$ image 2, <u>41</u>	$\backslash$ partial 62	
$\backslash$ Chi 2, <u>22</u>		$\backslash$ in 44	$\backslash$ plusmn 3, <u>63</u>	
$\backslash$ clubs 2, <u>23</u>		$\backslash$ infin 2, <u>42</u>	$\backslash$ pm 63	
$\backslash$ clubsuit 23		$\backslash$ infty 42	Q	
$\backslash$ cnums 2, <u>24</u>		$\backslash$ Iota 2, <u>43</u>	$\backslash$ Q 1, <u>12</u>	
$\backslash$ Complex 2, <u>25</u>		$\backslash$ isin 2, <u>44</u>	R	
$\backslash$ coppa 80		K		$\backslash$ R 2, <u>13</u>
$\backslash$ Cup 30		$\backslash$ Kappa 2, <u>45</u>	$\backslash$ rang 1, <u>5</u>	
D		L		$\backslash$ rangle 5
$\backslash$ Dagger 2, <u>26</u>		$\backslash$ land 18	$\backslash$ Rarr 3, <u>65</u>	
$\backslash$ Darr 1, <u>3</u>		$\backslash$ lang 1, <u>4</u>	$\backslash$ rArr 3, <u>66</u>	
$\backslash$ dArr 1, <u>2</u>		$\backslash$ langle 4	$\backslash$ rarr 3, <u>64</u>	
$\backslash$ darr 1, <u>1</u>		$\backslash$ Larr 3, <u>47</u>	$\backslash$ Re 67	
$\backslash$ ddagger 26		$\backslash$ lArr 3, <u>48</u>	$\backslash$ real 3, <u>67</u>	
$\backslash$ def 61		$\backslash$ larr 3, <u>46</u>	$\backslash$ Reals 3, <u>69</u>	
$\backslash$ diamonds 2, <u>27</u>		$\backslash$ le 3, <u>49</u>	$\backslash$ reals 3, <u>68</u>	
$\backslash$ diamondsuit 27		$\backslash$ Leftarrow 47, 48	$\backslash$ renewcommand 10, 18,	
$\backslash$ Doteq 2, <u>28</u>		$\backslash$ leftarrow 46	28, 29, 30, 31, 35,	
$\backslash$ doteqdot 28		$\backslash$ Leftrightarrow 37, 39, 51, 52	36, 49, 55, 57, 62, 70	
$\backslash$ doublecap 2, <u>29</u>		$\backslash$ leftrightharrow 38, 50	$\backslash$ restriction 3, <u>70</u>	
$\backslash$ doublecup 2, <u>30</u>		$\backslash$ leq 49	$\backslash$ Rho 3, <u>71</u>	
$\backslash$ Downarrow 2, 3		$\backslash$ let 60	$\backslash$ Rightarrow 65, 66	
$\backslash$ downarrow 1		$\backslash$ lor 61	$\backslash$ rightarrow 64	
E		$\backslash$ Lrarr 3, <u>51</u>	S	
$\backslash$ else 61		$\backslash$ lRarr 3, <u>52</u>	$\backslash$ S 73	
$\backslash$ empty 2, <u>31</u>		$\backslash$ lrarr 3, <u>50</u>	$\backslash$ sdot 3, <u>72</u>	
$\backslash$ emptyset 31, <u>57</u>		M		$\backslash$ sect 3, <u>73</u>
$\backslash$ Epsilon 2, <u>32</u>		$\backslash$ mathbb 9, 10, 11, 12, 13,	$\backslash$ spades 3, <u>74</u>	
$\backslash$ Eta 2, <u>33</u>		14, 24, 25, 54, 68, 69	$\backslash$ spadesuit 74	
$\backslash$ exist 2, <u>34</u>		$\backslash$ mathrm 17, 20, 22,	$\backslash$ sub 3, <u>75</u>	
$\backslash$ exists 34		32, 33, 43, 45, 53,	$\backslash$ sube 3, <u>76</u>	
$\backslash$ expandafter 61		56, 58, 59, 71, 78, 82	$\backslash$ subset 75	
F		$\backslash$ mbox 80	$\backslash$ subsetq 76	
$\backslash$ fi 61		$\backslash$ Mu 3, <u>53</u>	$\backslash$ supe 3, <u>77</u>	
G		N		$\backslash$ supsetq 77
$\backslash$ ge 2, <u>35</u>		$\backslash$ N 1, <u>11</u>	T	
$\backslash$ geq 35		$\backslash$ natnums 3, <u>54</u>	$\backslash$ Tau 3, <u>78</u>	
$\backslash$ ggg 36		$\backslash$ ne 3, <u>55</u>	$\backslash$ thetasym 3, <u>79</u>	
$\backslash$ gggtr 2, <u>36</u>		$\backslash$ neq 55	U	
H		$\backslash$ Nu 3, <u>56</u>	$\backslash$ Uarr 1, <u>8</u>	
$\backslash$ H 1, <u>10</u>			$\backslash$ uArr 1, <u>7</u>	
$\backslash$ hAar 2, <u>37</u>			$\backslash$ uarr 1, <u>6</u>	

$\Uparrow$		7, 8	ϑ		79	Z	
$\uparrow$		6				$\mathbb{Z}$	 2, <u>14</u>
$\upharpoonright$		70	W			$\mathbb{Z}$	 4, <u>82</u>
V			$\wp$		4, <u>81</u>		
φ		4, <u>80</u>	$\wp$		81		

A List of all commands supported

Section A lists all commands allowed by MediaWiki. This includes the commands provided by this package but also selected commands provided by other packages (cf. Section 3).

It was generated automatically using a script that processes a JSON file that is part of the MediaWiki Math extension.

A.1 Group big_literals

$\Big$ is rendered as $\Big($

$\Bigg$ is rendered as $\Bigg($

$\Bigl$ is rendered as $\Bigl($

$\Biggr$ is rendered as $\Biggr($

$\Bigl$ is rendered as $\Bigl($

$\Bigr$ is rendered as $\Bigr($

$\big$ is rendered as $\big($

$\bigg$ is rendered as $\bigg($

$\bigl$ is rendered as $\bigl($

$\bigr$ is rendered as $\bigr($

$\bigl$ is rendered as $\bigl($

$\bigr$ is rendered as $\bigr($

A.2 Group box_functions

$\hbox$ is rendered as x

`\mbox` is rendered as x

`\text` is rendered as x

`\vbox` is rendered as x

A.3 Group `color_function`

`\color` is rendered as *red*

`\pagecolor` is not rendered.

A.4 Group `declh_function`

`\bf` is rendered as

`\cal` is rendered as $\mathcal{x}$

`\it` is rendered as *x*

`\rm` is rendered as x

A.5 Group `definecolor_function`

`\definecolor` is rendered as

A.6 Group `fun_ar1`

`\acute` is rendered as $\acute{x}$

`\bar` is rendered as $\bar{x}$

`\bcancel` is rendered as $\cancel{x}$

`\bmod` is rendered as $\bmod x$

`\boldsymbol` is rendered as $\boldsymbol{x}$

`\breve` is rendered as $\breve{x}$

`\cancel` is rendered as $\cancel{x}$

`\check` is rendered as $\check{x}$

`\ddot` is rendered as $\ddot{x}$

`\dot` is rendered as $\dot{x}$

`\emph` is rendered as *x*

`\grave` is rendered as $\grave{x}$

`\hat` is rendered as $\hat{x}$

`\hphantom` is rendered as x

`\mathcal` is rendered as $\mathcal{x}$

`\mathclose` is rendered as x

`\mathfrak` is rendered as $\mathfrak{x}$

$\backslash\mathrm{it}$ is rendered as x
 $\backslash\mathrm{open}$ is rendered as x
 $\backslash\mathrm{hord}$ is rendered as x
 $\backslash\mathrm{punct}$ is rendered as x
 $\backslash\mathrm{sf}$ is rendered as x
 $\backslash\mathrm{tt}$ is rendered as x
 $\backslash\mathrm{overleftarrow{}}$ is rendered as $\overleftarrow{x}$
 $\backslash\mathrm{overleftrightharpoonrightarrow}$ is rendered as $\overleftrightharpoonrightarrow x$
 $\backslash\mathrm{overline{}}$ is rendered as $\overline{x}$
 $\backslash\mathrm{overrightarrow{}}$ is rendered as $\overrightarrow{x}$
 $\backslash\mathrm{phantom}$ is rendered as x
 $\backslash\mathrm{pmod}$ is rendered as $x \pmod{x}$
 $\backslash\mathrm{sqrt}$ is rendered as $\sqrt{x}$
 $\backslash\mathrm{textbf}$ is rendered as $\mathbf{x}$
 $\backslash\mathrm{textit}$ is rendered as x
 $\backslash\mathrm{textrm}$ is rendered as x
 $\backslash\mathrm{textsf}$ is rendered as x
 $\backslash\mathrm{texttt}$ is rendered as x
 $\backslash\mathrm{tilde}$ is rendered as $\tilde{x}$
 $\backslash\mathrm{underline}$ is rendered as $\underline{x}$
 $\backslash\mathrm{vec}$ is rendered as $\vec{x}$
 $\backslash\mathrm{vphantom}$ is rendered as x
 $\backslash\mathrm{widehat{}}$ is rendered as $\widehat{x}$
 $\backslash\mathrm{widetilde{}}$ is rendered as $\widetilde{x}$
 $\backslash\mathrm{xcancel}$ is rendered as $\cancel{x}$

A.7 Group `fun_ar1nb`

$\backslash\mathrm{hbb}$ is rendered as $\mathbb{x}$
 $\backslash\mathrm{hbf}$ is rendered as $\mathbf{x}$
 $\backslash\mathrm{hbin}$ is rendered as x
 $\backslash\mathrm{hop}$ is rendered as x
 $\backslash\mathrm{hrel}$ is rendered as x
 $\backslash\mathrm{hrm}$ is rendered as x
 $\backslash\mathrm{operatorname{}}$ is rendered as x

`\overarc` is rendered as $5.0\text{pt}\widehat{x}$

`\overbrace` is rendered as $\overbrace{x}$

`\underbrace` is rendered as $\underbrace{x}$

`\xleftarrow` is rendered as $\xleftarrow{x}$

`\xrightarrow` is rendered as $\xrightarrow{x}$

A.8 Group `fun_ar1opt`

`\sqrt` is rendered as $\sqrt{x}$

`\xleftarrow` is rendered as $\xleftarrow{x}$

`\xrightarrow` is rendered as $\xrightarrow{x}$

A.9 Group `fun_ar2`

`\binom` applied on xx is rendered as $\binom{x}{x}$

`\cancelto` applied on xx is rendered as $x\cancel{\rightarrow}^x$

`\cfrac` applied on xx is rendered as $\frac{x}{x}$

`\dbinom` applied on xx is rendered as $\dbinom{x}{x}$

`\dfrac` applied on xx is rendered as $\dfrac{x}{x}$

`\frac` applied on xx is rendered as $\frac{x}{x}$

`\overset` applied on xx is rendered as $\overset{x}{x}$

`\stackrel` applied on xx is rendered as $\stackrel{x}{x}$

`\tbinom` applied on xx is rendered as $\tbinom{x}{x}$

`\tfrac` applied on xx is rendered as $\tfrac{x}{x}$

`\underset` applied on xx is rendered as $\underset{x}{x}$

A.10 Group `fun_ar2nb`

`\sideset` applied on $\sum_{13}^{24}$ is rendered as $\sum_{13}^{24}$

A.11 Group `fun_infix`

`\atop` applied on x, y is rendered as $\atop{x}{y}$

`\choose` applied on x, y is rendered as $\binom{x}{y}$

`\over` applied on x, y is rendered as $\frac{x}{y}$

A.12 Group `fun_mhchem`

`\ce` is rendered as x

A.13 Group `hline_function`

`\hline` applied in a table is rendered as $\underline{x_{11} \quad x_{12}}$

A.14 Group `latex_function_names`

`\Pr` is rendered as $\Pr$

`\arccos` is rendered as $\arccos$

`\arcsin` is rendered as $\arcsin$

`\arctan` is rendered as $\arctan$

`\arg` is rendered as $\arg$

`\cos` is rendered as $\cos$

`\cosh` is rendered as $\cosh$

`\cot` is rendered as $\cot$

`\coth` is rendered as $\coth$

`\csc` is rendered as $\csc$

`\deg` is rendered as $\deg$

`\det` is rendered as $\det$

`\dim` is rendered as $\dim$

`\exp` is rendered as $\exp$

`\gcd` is rendered as $\gcd$

`\hom` is rendered as $\hom$

`\inf` is rendered as $\inf$

`\ker` is rendered as $\ker$

`\lg` is rendered as $\lg$

`\lim` is rendered as $\lim$

`\liminf` is rendered as $\liminf$

`\limsup` is rendered as $\limsup$

`\ln` is rendered as $\ln$

`\log` is rendered as $\log$

`\max` is rendered as $\max$

`\min` is rendered as $\min$

`\sec` is rendered as $\sec$

`\sin` is rendered as $\sin$

`\sinh` is rendered as $\sinh$

`\sup` is rendered as $\sup$

`\tan` is rendered as $\tan$

`\tanh` is rendered as $\tanh$

A.15 Group `left_function`

`\left` is rendered as $($

A.16 Group `mediawiki_function_names`

`\arccot` is rendered as $\operatorname{arccot} y$

`\arccsc` is rendered as $\operatorname{arccsc} y$

`\arcsec` is rendered as $\operatorname{arcsec} y$

`\sen` is rendered as $\operatorname{sen} y$

`\sgn` is rendered as $\operatorname{sgn} y$

A.17 Group `mhchem_bond`

`\bond` is rendered as --

A.18 Group `mhchem_macro_1p`

`\ce` is rendered as x

`\mathbf` is rendered as $\mathbf{x}$

A.19 Group `mhchem_macro_2p`

`\frac` applied on xx is rendered as $\frac{x}{x}$

`\overset` applied on xx is rendered as $\overset{x}{x}$

`\underset` applied on xx is rendered as $\underset{x}{x}$

A.20 Group `mhchem_macro_2pc`

`\color` is rendered as red

A.21 Group `mhchem_macro_2pu`

`\underbrace` is rendered as $\underbrace{x}$

A.22 Group mhchem_single_macro

`\Alpha` is rendered as A

`\Beta` is rendered as B

`\Chi` is rendered as X

`\Delta` is rendered as Δ

`\Epsilon` is rendered as E

`\Eta` is rendered as H

`\Gamma` is rendered as Γ

`\Iota` is rendered as I

`\Kappa` is rendered as K

`\Lambda` is rendered as Λ

`\Mu` is rendered as M

`\Nu` is rendered as N

`\Omega` is rendered as Ω

`\Omicron` is rendered as O

`\Phi` is rendered as Φ

`\Pi` is rendered as Π

`\Psi` is rendered as Ψ

`\Rho` is rendered as P

`\Sigma` is rendered as Σ

`\Tau` is rendered as T

`\Theta` is rendered as Θ

`\Upsilon` is rendered as Υ

`\Zeta` is rendered as Z

`\alpha` is rendered as α

`\approx` is rendered as $\approx$

`\beta` is rendered as β

`\chi` is rendered as χ

`\circ` is rendered as $\circ$

`\delta` is rendered as δ

`\epsilon` is rendered as ϵ

`\eta` is rendered as η

`\gamma` is rendered as γ

$\backslash$ iota is rendered as ι
 $\backslash$ kappa is rendered as κ
 $\backslash$ lambda is rendered as λ
 $\backslash$ mu is rendered as μ
 $\backslash$ nu is rendered as ν
 $\backslash$ omega is rendered as ω
 $\backslash$ omicron is rendered as $\omicron$
 $\backslash$ phi is rendered as φ
 $\backslash$ pi is rendered as π
 $\backslash$ pm is rendered as $\pm$
 $\backslash$ psi is rendered as ψ
 $\backslash$ rho is rendered as ρ
 $\backslash$ sigma is rendered as σ
 $\backslash$ tau is rendered as τ
 $\backslash$ theta is rendered as ϑ
 $\backslash$ upsilon is rendered as υ
 $\backslash$ varepsilon is rendered as ϵ
 $\backslash$ varkappa is rendered as $\varkappa$
 $\backslash$ varphi is rendered as φ
 $\backslash$ varpi is rendered as ϖ
 $\backslash$ varrho is rendered as ϱ
 $\backslash$ varsigma is rendered as ς
 $\backslash$ vartheta is rendered as ϑ
 $\backslash$ zeta is rendered as ζ

A.23 Group nullary_macro

$\backslash$ And is rendered as ς
 $\backslash$ Bbbk is rendered as $\mathbb{k}$
 $\backslash$ Box is rendered as $\square$
 $\backslash$ Bumpeq is rendered as $\approx$
 $\backslash$ Cap is rendered as $\mathfrak{m}$
 $\backslash$ Cup is rendered as $\mathfrak{w}$
 $\backslash$ Delta is rendered as Δ
 $\backslash$ Diamond is rendered as $\diamond$

$\backslash\text{Finv}$ is rendered as $\dashv$
 $\backslash\text{Game}$ is rendered as $\oslash$
 $\backslash\text{Gamma}$ is rendered as Γ
 $\backslash\text{Im}$ is rendered as $\Im$
 $\backslash\text{Lambda}$ is rendered as Λ
 $\backslash\text{Leftarrow}$ is rendered as $\Leftarrow$
 $\backslash\text{Leftrightarrow}$ is rendered as $\Leftrightarrow$
 $\backslash\text{Lleftarrow}$ is rendered as $\Lleftarrow$
 $\backslash\text{Longleftarrow}$ is rendered as $\Longleftarrow$
 $\backslash\text{Longleftrightarrow}$ is rendered as $\Longleftrightarrow$
 $\backslash\text{Longrightarrow}$ is rendered as $\Longrightarrow$
 $\backslash\text{Lsh}$ is rendered as $\lsh$
 $\backslash\text{Omega}$ is rendered as Ω
 $\backslash\text{P}$ is rendered as $\P$
 $\backslash\text{Phi}$ is rendered as Φ
 $\backslash\text{Pi}$ is rendered as Π
 $\backslash\text{Psi}$ is rendered as Ψ
 $\backslash\text{Re}$ is rendered as $\Re$
 $\backslash\text{Rrightarrow}$ is rendered as $\Rightarrow$
 $\backslash\text{Rrightarrow}$ is rendered as $\Rrightarrow$
 $\backslash\text{Rsh}$ is rendered as $\rsh$
 $\backslash\text{S}$ is rendered as $\S$
 $\backslash\text{Sigma}$ is rendered as Σ
 $\backslash\text{Subset}$ is rendered as $\Subset$
 $\backslash\text{Supset}$ is rendered as $\Supset$
 $\backslash\text{Theta}$ is rendered as Θ
 $\backslash\text{Upsilon}$ is rendered as Υ
 $\backslash\text{Vdash}$ is rendered as $\Vdash$
 $\backslash\text{Vvdash}$ is rendered as $\Vvdash$
 $\backslash\text{Xi}$ is rendered as Ξ
 $\backslash\text{aleph}$ is rendered as $\aleph$
 $\backslash\text{alpha}$ is rendered as α
 $\backslash\text{amalg}$ is rendered as $\amalg$

`\angle` is rendered as $\angle$
`\approx` is rendered as $\approx$
`\approxeq` is rendered as $\cong$
`\ast` is rendered as $*$
`\asymp` is rendered as $\asymp$
`\backepsilon` is rendered as ϵ
`\backprime` is rendered as $\backprime$
`\backsimeq` is rendered as $\backsimeq$
`\backsimeq` is rendered as $\backsimeq$
`\barwedge` is rendered as $\barwedge$
`\because` is rendered as $\because$
`\beta` is rendered as β
`\beth` is rendered as $\beth$
`\between` is rendered as $\between$
`\bigcap` is rendered as $\bigcap$
`\bigcirc` is rendered as $\bigcirc$
`\bigcup` is rendered as $\bigcup$
`\bigodot` is rendered as $\bigodot$
`\bigoplus` is rendered as $\bigoplus$
`\bigotimes` is rendered as $\bigotimes$
`\bigsqcup` is rendered as $\bigsqcup$
`\bigstar` is rendered as $\bigstar$
`\bigtriangledown` is rendered as $\bigtriangledown$
`\bigtriangleup` is rendered as $\bigtriangleup$
`\biguplus` is rendered as $\biguplus$
`\bigvee` is rendered as $\bigvee$
`\bigwedge` is rendered as $\bigwedge$
`\blacklozenge` is rendered as $\blacklozenge$
`\blacksquare` is rendered as $\blacksquare$
`\blacktriangle` is rendered as $\blacktriangle$
`\blacktriangledown` is rendered as $\blacktriangledown$
`\blacktriangleleft` is rendered as $\blacktriangleleft$
`\blacktriangleright` is rendered as $\blacktriangleright$

`\bot` is rendered as $\perp$
`\bowtie` is rendered as $\bowtie$
`\boxdot` is rendered as $\boxdot$
`\boxminus` is rendered as $\boxminus$
`\boxplus` is rendered as $\boxplus$
`\boxtimes` is rendered as $\boxtimes$
`\bullet` is rendered as $\bullet$
`\bumpeq` is rendered as $\bumpeq$
`\cap` is rendered as $\cap$
`\cdot` is rendered as $\cdot$
`\cdots` is rendered as $\cdots$
`\centerdot` is rendered as $\cdot$
`\checkmark` is rendered as $\checkmark$
`\chi` is rendered as χ
`\circ` is rendered as $\circ$
`\circeq` is rendered as $\overset{\circ}{=}$
`\circlearrowleft` is rendered as $\circlearrowleft$
`\circlearrowright` is rendered as $\circlearrowright$
`\circledS` is rendered as $\textcircled{\text{S}}$
`\circledast` is rendered as $\textcircled{\ast}$
`\circledcirc` is rendered as $\textcircled{\circ}$
`\circleddash` is rendered as $\textcircled{-}$
`\clubsuit` is rendered as $\clubsuit$
`\colon` is rendered as $:$
`\complement` is rendered as $\complement$
`\cong` is rendered as $\cong$
`\coprod` is rendered as $\coprod$
`\cup` is rendered as $\cup$
`\curlyeqprec` is rendered as $\curlyeqprec$
`\curlyeqsucc` is rendered as $\curlyeqsucc$
`\curlyvee` is rendered as $\curlyvee$
`\curlywedge` is rendered as $\curlywedge$
`\curvearrowleft` is rendered as $\curvearrowleft$

`\curvearrowright` is rendered as $\curvearrowright$
`\dagger` is rendered as $\dagger$
`\daleth` is rendered as $\daleth$
`\dashv` is rendered as $\dashv$
`\ddagger` is rendered as $\ddagger$
`\ddots` is rendered as $\ddots$
`\delta` is rendered as δ
`\diagdown` is rendered as $\diagdown$
`\diagup` is rendered as $\diagup$
`\diamond` is rendered as $\diamond$
`\diamondsuit` is rendered as $\diamondsuit$
`\digamma` is rendered as $\digamma$
`\displaystyle` is rendered as
$$`\div` is rendered as $\div$
`\divideontimes` is rendered as $\divideontimes$
`\doteq` is rendered as $\doteq$
`\doteqdot` is rendered as $\doteqdot$
`\dotplus` is rendered as $\dotplus$
`\dots` is rendered as $\dots$
`\dotsb` is rendered as $\dotsb$
`\dotsc` is rendered as $\dotsc$
`\dotsi` is rendered as $\dotsi$
`\dotsm` is rendered as $\dotsm$
`\dotso` is rendered as $\dotso$
`\doublebarwedge` is rendered as $\doublebarwedge$
`\downdownarrows` is rendered as $\downdownarrows$
`\downharpoonleft` is rendered as $\downharpoonleft$
`\downharpoonright` is rendered as $\downharpoonright$
`\ell` is rendered as ℓ
`\emptyset` is rendered as $\emptyset$
`\epsilon` is rendered as ϵ
`\eqcirc` is rendered as $\eqcirc$
`\eqsim` is rendered as $\eqsim$$$

`\eqslantgtr` is rendered as $\gtrless$
`\eqslantless` is rendered as $\lessgtr$
`\equiv` is rendered as $\equiv$
`\eta` is rendered as η
`\eth` is rendered as $\eth$
`\exists` is rendered as $\exists$
`\fallingdotseq` is rendered as $\fallingdotseq$
`\flat` is rendered as $\flat$
`\forall` is rendered as $\forall$
`\frown` is rendered as $\frown$
`\gamma` is rendered as γ
`\geq` is rendered as $\geq$
`\geqq` is rendered as $\geqq$
`\geqslant` is rendered as $\geqslant$
`\gets` is rendered as $\leftarrow$
`\gg` is rendered as $\gg$
`\ggg` is rendered as $\ggg$
`\gimel` is rendered as $\gimel$
`\gnapprox` is rendered as $\gtrapprox$
`\gneq` is rendered as $\gtrneq$
`\gneqq` is rendered as $\gtrneqq$
`\gnsim` is rendered as $\gtrsim$
`\gtrapprox` is rendered as $\gtrapprox$
`\gtrdot` is rendered as $\gtrdot$
`\gtreqless` is rendered as $\gtreqless$
`\gtreqqless` is rendered as $\gtreqqless$
`\gtrless` is rendered as $\gtrless$
`\gtrsim` is rendered as $\gtrsim$
`\gvertneqq` is rendered as $\gvertneqq$
`\hbar` is rendered as $\hbar$
`\heartsuit` is rendered as $\heartsuit$
`\hookleftarrow` is rendered as $\hookleftarrow$
`\hookrightarrow` is rendered as $\hookrightarrow$

`\hslash` is rendered as $\hbar$
`\iff` is rendered as $\iff$
`\iiint` is rendered as $\iiint$
`\iiint` is rendered as $\iiint$
`\iint` is rendered as $\iint$
`\imath` is rendered as $\imath$
`\implies` is rendered as $\implies$
`\in` is rendered as $\in$
`\infty` is rendered as ∞
`\injl` is rendered as $\injl$
`\int` is rendered as $\int$
`\intBar` is rendered as $\intBar$
`\intbar` is rendered as $\intbar$
`\intercal` is rendered as $\intercal$
`\iota` is rendered as ι
`\jmath` is rendered as $\jmath$
`\kappa` is rendered as κ
`\lVert` is rendered as $\lVert$
`\lambda` is rendered as λ
`\land` is rendered as $\land$
`\ldots` is rendered as $\ldots$
`\leftarrow` is rendered as $\leftarrow$
`\leftarrowtail` is rendered as $\leftarrowtail$
`\leftharpoondown` is rendered as $\leftharpoondown$
`\leftharpoonup` is rendered as $\leftharpoonup$
`\leftleftarrows` is rendered as $\leftleftarrows$
`\leftrightarrows` is rendered as $\leftrightarrows$
`\leftrightharpoons` is rendered as $\leftrightharpoons$
`\leftrightsquigarrow` is rendered as $\leftrightsquigarrow$
`\leftthreetimes` is rendered as $\leftthreetimes$
`\leq` is rendered as $\leq$
`\leqq` is rendered as $\leqq$

`\leqslant` is rendered as $\leqslant$
`\lessapprox` is rendered as $\lessapprox$
`\lessdot` is rendered as $\lessdot$
`\lesseqgtr` is rendered as $\lesseqgtr$
`\lesseqqgtr` is rendered as $\lesseqqgtr$
`\lessgtr` is rendered as $\lessgtr$
`\lesssim` is rendered as $\lesssim$
`\limits` is rendered for example as $\bigcap_a^b$
`\ll` is rendered as $\ll$
`\lll` is rendered as $\lll$
`\lnapprox` is rendered as $\lnapprox$
`\lneq` is rendered as $\lneq$
`\lneqq` is rendered as $\lneqq$
`\lnot` is rendered as $\lnot$
`\lnsim` is rendered as $\lnsim$
`\longleftarrow` is rendered as $\longleftarrow$
`\longlefttrightarrow` is rendered as $\longleftrightarrow$
`\longmapsto` is rendered as $\longmapsto$
`\longrightarrow` is rendered as $\longrightarrow$
`\looparrowleft` is rendered as $\looparrowleft$
`\looparrowright` is rendered as $\looparrowright$
`\lor` is rendered as $\lor$
`\lozenge` is rendered as $\lozenge$
`\ltimes` is rendered as $\ltimes$
`\lvertneqq` is rendered as $\lvertneqq$
`\mapsto` is rendered as $\mapsto$
`\measuredangle` is rendered as $\measuredangle$
`\mho` is rendered as $\mho$
`\mid` is rendered as $\mid$
`\mod` is rendered as $\bmod$
`\models` is rendered as $\models$
`\mp` is rendered as $\mp$
`\mu` is rendered as μ

`\multimap` is rendered as $\multimap$
`\nLeftarrow` is rendered as $\nLeftarrow$
`\nLeftrightarrow` is rendered as $\nLeftrightarrow$
`\nRightarrow` is rendered as $\nRightarrow$
`\nVDash` is rendered as $\nVDash$
`\nVdash` is rendered as $\nVdash$
`\nabla` is rendered as ∇
`\natural` is rendered as $\natural$
`\ncong` is rendered as $\ncong$
`\nearrow` is rendered as $\nearrow$
`\neg` is rendered as $\neg$
`\neq` is rendered as $\neq$
`\nexists` is rendered as $\nexists$
`\ngeq` is rendered as $\ngeq$
`\ngeqq` is rendered as $\ngeqq$
`\ngeqslant` is rendered as $\ngeqslant$
`\ngtr` is rendered as $\ngtr$
`\ni` is rendered as $\ni$
`\nleftarrow` is rendered as $\nleftarrow$
`\nleftrightarrow` is rendered as $\nleftrightarrow$
`\nleq` is rendered as $\nleq$
`\nleqq` is rendered as $\nleqq$
`\nleqslant` is rendered as $\nleqslant$
`\nless` is rendered as $\nless$
`\nmid` is rendered as $\nmid$
`\nolimits` is rendered for example as $\cap_a^b$
`\not` is rendered as $\not$
`\notin` is rendered as $\notin$
`\nparallel` is rendered as $\nparallel$
`\nprec` is rendered as $\nprec$
`\npreceq` is rendered as $\npreceq$
`\rightarrow` is rendered as $\rightarrow$
`\nshortmid` is rendered as $\nshortmid$

`\nshortparallel` is rendered as $\nshortparallel$
`\nsim` is rendered as $\sim$
`\nsubseteq` is rendered as $\nsubseteq$
`\nsubseteqq` is rendered as $\nsubseteqq$
`\nsucc` is rendered as $\succ$
`\nsucceq` is rendered as $\succeq$
`\nsupseteq` is rendered as $\nsupseteq$
`\nsupseteqq` is rendered as $\nsupseteqq$
`\ntriangleleft` is rendered as $\ntriangleleft$
`\ntrianglelefteq` is rendered as $\ntrianglelefteq$
`\ntriangleright` is rendered as $\ntriangleright$
`\ntrianglerighteq` is rendered as $\ntrianglerighteq$
`\nu` is rendered as ν
`\nvDash` is rendered as $\nvDash$
`\nvDash` is rendered as $\nVdash$
`\nwarrow` is rendered as $\nwarrow$
`\odot` is rendered as $\odot$
`\oiint` is rendered as $\oiint$
`\oiiint` is rendered as $\oiiint$
`\oint` is rendered as $\oint$
`\ointctrclockwise` is rendered as $\ointctrclockwise$
`\omega` is rendered as ω
`\ominus` is rendered as $\ominus$
`\oplus` is rendered as $\oplus$
`\oslash` is rendered as $\oslash$
`\otimes` is rendered as $\otimes$
`\parallel` is rendered as $\parallel$
`\partial` is rendered as ∂
`\perp` is rendered as $\perp$
`\phi` is rendered as ϕ
`\pi` is rendered as π
`\pitchfork` is rendered as $\pitchfork$
`\pm` is rendered as $\pm$

`\prec` is rendered as $\prec$
`\precapprox` is rendered as $\approx$
`\preccurlyeq` is rendered as $\preccurlyeq$
`\preceq` is rendered as $\preceq$
`\precnapprox` is rendered as $\approx$
`\precneqq` is rendered as $\neq$
`\precnsim` is rendered as $\sim$
`\precsim` is rendered as $\lesssim$
`\prime` is rendered as $'$
`\prod` is rendered as $\prod$
`\projlim` is rendered as proj lim
`\propto` is rendered as $\propto$
`\psi` is rendered as ψ
`\quad` is rendered as $\quad$
`\quad` is rendered as $\quad$
`\rVert` is rendered as $\parallel$
`\rho` is rendered as ρ
`\rightarrow` is rendered as $\rightarrow$
`\rightarrowtail` is rendered as $\rightarrowtail$
`\rightharpoondown` is rendered as $\searrow$
`\rightharpoonup` is rendered as $\nearrow$
`\rightleftarrows` is rendered as $\rightleftarrows$
`\rightrightarrows` is rendered as $\rightrightarrows$
`\rightsquigarrow` is rendered as $\rightsquigarrow$
`\rightthreetimes` is rendered as $\rightthreetimes$
`\risingdotseq` is rendered as $\dot{=}$
`\rtimes` is rendered as $\rtimes$
`\scriptscriptstyle` is rendered as $\scriptscriptstyle$
`\scriptstyle` is rendered as $\scriptstyle$
`\searrow` is rendered as $\searrow$
`\setminus` is rendered as $\setminus$
`\sharp` is rendered as $\sharp$
`\shortmid` is rendered as $\shortmid$

`\shortparallel` is rendered as $\parallel$
`\sigma` is rendered as σ
`\sim` is rendered as $\sim$
`\simeq` is rendered as $\simeq$
`\smallfrown` is rendered as $\frown$
`\smallsetminus` is rendered as $\setminus$
`\smallsmile` is rendered as $\smile$
`\smile` is rendered as $\smile$
`\spadesuit` is rendered as $\spadesuit$
`\sphericalangle` is rendered as $\sphericalangle$
`\sqcap` is rendered as $\sqcap$
`\sqcup` is rendered as $\sqcup$
`\sqsubset` is rendered as $\sqsubset$
`\sqsubseteq` is rendered as $\sqsubseteq$
`\sqsupset` is rendered as $\sqsupset$
`\sqsupseteq` is rendered as $\sqsupseteq$
`\square` is rendered as $\square$
`\star` is rendered as $\star$
`\subset` is rendered as $\subset$
`\subseteq` is rendered as $\subseteq$
`\subseteqq` is rendered as $\subseteqq$
`\subsetneq` is rendered as $\subsetneq$
`\subsetneqq` is rendered as $\subsetneqq$
`\succ` is rendered as $\succ$
`\succapprox` is rendered as $\succapprox$
`\succcurlyeq` is rendered as $\succcurlyeq$
`\succeq` is rendered as $\succeq$
`\succnapprox` is rendered as $\succnapprox$
`\succneqq` is rendered as $\succneqq$
`\succnsim` is rendered as $\succnsim$
`\succsim` is rendered as $\succsim$
`\sum` is rendered as $\sum$
`\supset` is rendered as $\supset$

`\supseteq` is rendered as $\supseteq$
`\supseteqq` is rendered as $\supseteqq$
`\supsetneq` is rendered as $\supsetneq$
`\supsetneqq` is rendered as $\supsetneqq$
`\surd` is rendered as $\surd$
`\swarrow` is rendered as $\swarrow$
`\tau` is rendered as τ
`\textstyle` is rendered as $\textstyle$
`\therefore` is rendered as $\therefore$
`\theta` is rendered as θ
`\thickapprox` is rendered as $\thickapprox$
`\thicksim` is rendered as $\thicksim$
`\times` is rendered as $\times$
`\rightarrow` is rendered as $\rightarrow$
`\top` is rendered as $\top$
`\triangle` is rendered as $\triangle$
`\triangledown` is rendered as $\triangledown$
`\triangleleft` is rendered as $\triangleleft$
`\trianglelefteq` is rendered as $\trianglelefteq$
`\triangleq` is rendered as $\triangleq$
`\triangleright` is rendered as $\triangleright$
`\trianglerighteq` is rendered as $\trianglerighteq$
`\upharpoonleft` is rendered as $\upharpoonleft$
`\upharpoonright` is rendered as $\upharpoonright$
`\uplus` is rendered as $\uplus$
`\upsilon` is rendered as υ
`\upuparrows` is rendered as $\upuparrows$
`\vDash` is rendered as $\vDash$
`\varDelta` is rendered as $\varDelta$
`\varGamma` is rendered as $\varGamma$
`\varLambda` is rendered as $\varLambda$
`\varOmega` is rendered as $\varOmega$
`\varPhi` is rendered as $\varPhi$

$\backslash\text{varPi}$ is rendered as Π
 $\backslash\text{varSigma}$ is rendered as Σ
 $\backslash\text{varTheta}$ is rendered as Θ
 $\backslash\text{varUpsilon}$ is rendered as Υ
 $\backslash\text{varXi}$ is rendered as Ξ
 $\backslash\text{varepsilon}$ is rendered as ϵ
 $\backslash\text{varinjlim}$ is rendered as $\varinjlim$
 $\backslash\text{varkappa}$ is rendered as κ
 $\backslash\text{varliminf}$ is rendered as $\varliminf$
 $\backslash\text{varlimsup}$ is rendered as $\varlimsup$
 $\backslash\text{varnothing}$ is rendered as $\emptyset$
 $\backslash\text{varointclockwise}$ is rendered as $\oint$
 $\backslash\text{varphi}$ is rendered as φ
 $\backslash\text{varpi}$ is rendered as ϖ
 $\backslash\text{varprojlim}$ is rendered as $\varprojlim$
 $\backslash\text{varpropto}$ is rendered as $\propto$
 $\backslash\text{varrho}$ is rendered as ϱ
 $\backslash\text{varsigma}$ is rendered as ς
 $\backslash\text{varsubsetneq}$ is rendered as $\subsetneq$
 $\backslash\text{varsubsetneqq}$ is rendered as $\subsetneqq$
 $\backslash\text{varsupsetneq}$ is rendered as $\supsetneq$
 $\backslash\text{varsupsetneqq}$ is rendered as $\supsetneqq$
 $\backslash\text{vartheta}$ is rendered as ϑ
 $\backslash\text{vartriangle}$ is rendered as $\triangle$
 $\backslash\text{vartriangleleft}$ is rendered as $\triangleleft$
 $\backslash\text{vartriangleright}$ is rendered as $\triangleright$
 $\backslash\text{vdash}$ is rendered as $\vdash$
 $\backslash\text{vdots}$ is rendered as $\vdots$
 $\backslash\text{vee}$ is rendered as $\vee$
 $\backslash\text{veebar}$ is rendered as $\underline{\vee}$
 $\backslash\text{vline}$ is rendered as $|$
 $\backslash\text{wedge}$ is rendered as $\wedge$
 $\backslash\text{wp}$ is rendered as $\wp$

`\wr` is rendered as $\wr$

`\xi` is rendered as ξ

`\zeta` is rendered as ζ

A.24 Group `nullary_macro_in_mbox`

`\AA` is rendered as $\forall$

`\Coppa` is rendered as λ

`\Digamma` is rendered as $\P$

`\Koppa` is rendered as λ

`\Sampi` is rendered as ν

`\Stigma` is rendered as μ

`\coppa` is rendered as $\mathfrak{t}$

`\euro` is rendered as e

`\geneuro` is rendered as €

`\geneuronarrow` is rendered as €

`\geneurowide` is rendered as €

`\koppa` is rendered as θ

`\officialeguro` is rendered as e

`\sampi` is rendered as σ

`\stigma` is rendered as Σ

`\textvisiblespace` is rendered as ~

`\varstigma` is rendered as Υ

A.25 Group `other_delimiters1`

`\Downarrow` is rendered as $\Downarrow$

`\Uparrow` is rendered as $\Uparrow$

`\Updownarrow` is rendered as $\Updownarrow$

`\Vert` is rendered as $\|$

`\backslash` is rendered as $\backslash$

`\downarrow` is rendered as $\downarrow$

`\langle` is rendered as $\langle$

`\lbrace` is rendered as $\{$

`\lbrack` is rendered as $[$

`\lceil` is rendered as $\lceil$

$\lfloor$ is rendered as $\lfloor$
 $\llcorner$ is rendered as $\llcorner$
 $\lrcorner$ is rendered as $\lrcorner$
 $\rangle$ is rendered as $\rangle$
 $\rbrace$ is rendered as $\}$
 $\rbrack$ is rendered as $\}$
 $\rceil$ is rendered as $\rceil$
 $\rfloor$ is rendered as $\rfloor$
 $\rightleftharpoons$ is rendered as $\rightleftharpoons$
 $\twoheadleftarrow$ is rendered as $\twoheadleftarrow$
 $\twoheadrightarrow$ is rendered as $\twoheadrightarrow$
 $\ulcorner$ is rendered as $\ulcorner$
 $\uparrow$ is rendered as $\uparrow$
 $\updownarrow$ is rendered as $\updownarrow$
 $\urcorner$ is rendered as $\urcorner$
 $\vert$ is rendered as $\vert$

A.26 Group `other_delimiters2`

$\Darr$ is rendered as $\Downarrow$
 $\Uarr$ is rendered as $\Uparrow$
 $\dArr$ is rendered as $\Downarrow$
 $\darr$ is rendered as $\downarrow$
 $\lang$ is rendered as $\langle$
 $\rang$ is rendered as $\rangle$
 $\uArr$ is rendered as $\Uparrow$
 $\uarr$ is rendered as $\uparrow$

A.27 Group `right_function`

$\right$ is rendered as $)$